

E1-03 Candidate Methods

source: research/candidate_methods.md — v1.4.0 (2026-09-22)

Candidate methods (P01, protocol v1.0)

Research questions

- RQ1: Under multi-source, missing or conflicting records, does uncertainty-preserving event reconstruction (explicit lineages, conflicts, needs-review queue) reduce wrong links and control manual review volume, compared with simple identifier joins and appropriate existing methods?
- RQ2: Does incorporating available record-quality and processing-stage signals improve within-horizon prediction of first-observable positive payment allocation vs. no-payment closure vs. still-open, relative to aging-rule and basic baselines?

Both are hypotheses to be tested, not established improvements.

Module A candidates (reconstruction)

A0 - exact_id baseline

- Inputs: native previous_record_id plus confirmed mappings only.
- Decision: link when explicit reference exists and passes org/currency/time sanity checks.
- Failure condition: broken or circular references surface as conflicts.
- Cost: trivial. Source: standard deterministic join.

A1 - constrained_linkage candidate

- Candidate generation from business references, date windows and amount patterns within org + source system; never across orgs; never on payer/service token alone.
- Rules carry ID, input dependencies, action, conflict priority and provenance.
- Multi-candidate ambiguity is preserved as needs_review instead of forced picks.
- Confidence scores labeled UNVALIDATED unless calibrated.
- Difference vs A0: higher coverage with an explicit uncertainty tier and downstream conflict/amount consistency checks.

Module B candidates (prediction)

B0 - age/stage rule baseline

- Transparent rules or training-set empirical risk by aging bucket and known stage.
- Rules are not probabilities; empirical risk states its sample support.

B1 - discrete multinomial baseline

- Discrete-time competing-event model: per-day interval conditional probabilities (payment / close-no-payment / none) from t-time features only.
- $S(0)=1$; $F_payment(h)=\sum_{k \leq h} S(k-1) p_payment(k)$; F_close analogously; $S(h)=S(h-1) p_none(h)$; class probabilities sum to 1 at every step.

C1 - quality-augmented candidate

- Same modeling frame as B1 plus reconstruction-quality / missingness / conflict features computed strictly from data available at t.
- Explicitly a candidate to be tested; not claimed original in advance.

C2 (optional)

- Only if an independent reason appears (e.g. demonstrated non-linearity B1 cannot capture). Not added for appearance.

Evaluation

- Module A: precision/recall on adjudicable samples, wrong-merge/split counts, refusal ratio, coverage; unlabeled and unknown reported separately.
- Module B: log-loss, three-class Brier score, grouped calibration, class counts, refusal/censoring proportions; H-horizon metrics restricted to known-outcome or fully-covered cases with the selection disclosed.
- Ablations: drop quality features (C1 vs B1); use baseline reconstruction feeding B1.